



**EFY GROUP**  
YOURS SINCE 1989

**Facing problems in getting your favourite EFY Magazine in your area?**

**We can help you!**



**Electronics For You magazine** is available on all top news-stands/outlets in India. If you are still facing problems in getting EFY magazine in your area, please call Akhilesh Yadav on **+91-8800295612** or drop an email at [efycirc@efy.in](mailto:efycirc@efy.in). We will help you locate the nearest news-stand or arrange a copy at your doorstep.



South Asia's Most Popular Electronics Magazine

To become an agent of EFY Group publications, call Akhilesh Yadav on **+91-8800295612** or write at [efycirc@efy.in](mailto:efycirc@efy.in)

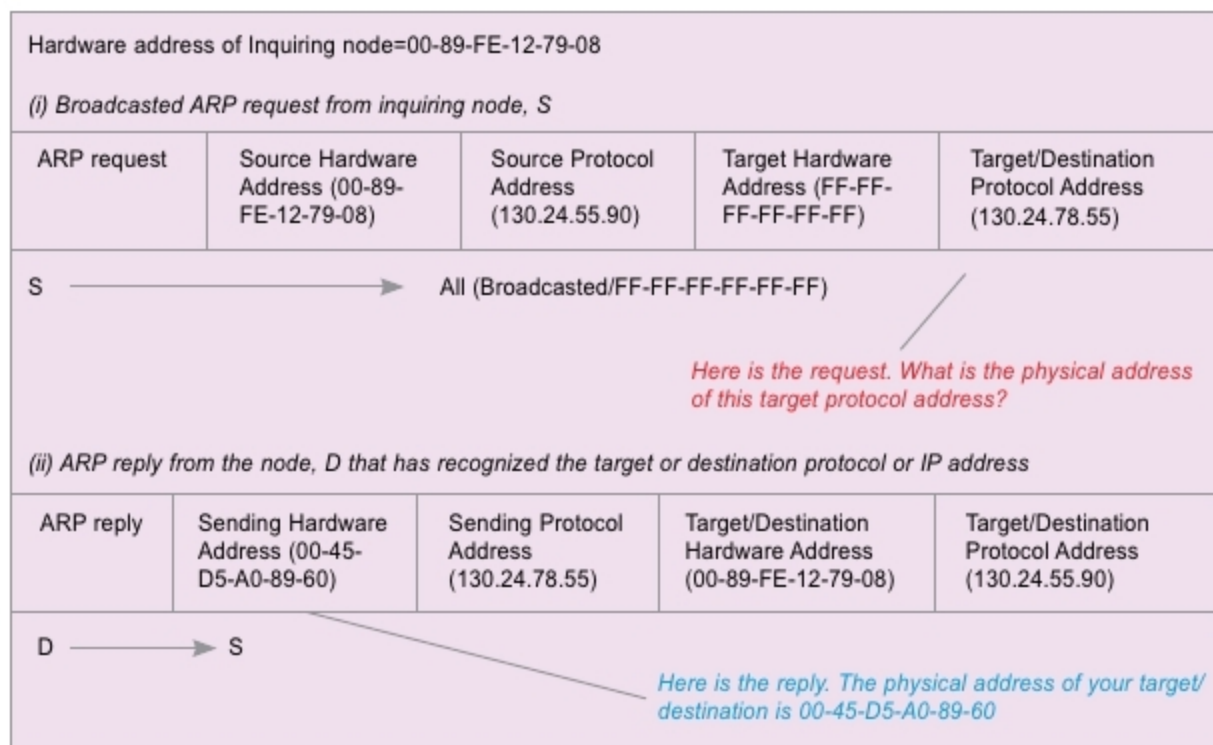


Fig. 3: ARP reply and request

data link and network layers (complete node coverage/node in OSI/ISO reference, which equals physical + data link + network layer). An example would be any IEEE802 series LAN to any X.25 WAN or MAN connection.

- By gateways that are used for connection networks of different protocols. A gateway covers all layers of a protocol. These are logically full protocol converters. An example would be a connection between an OSI/ISO network and a DNA network.

Repeaters, bridges and routers are used when interconnecting networks are of the same protocol. Gateways, repeaters, bridges and routers are all made of computing resources. In fact, these are all computers. Complexity of systems increases as one moves from repeaters to gateways. Bridges and routers can be defined as simple gateways. All these devices interconnect incompatible networks physically. The Internet does so logically for all networks globally.

## Other Internet protocols

**Address resolution protocol (ARP).** IP addresses are unique logical addresses, each assigned to a physical machine, system or node. An IP layer releases IP packets with IP logical addresses to the physical layer for the purpose of transport. The physical layer treats each IP packet as a data pack and encapsulates the pack into the physically-enabled transport packet commensurate to

the physical network. For example, if the physical transport network is Ethernet LAN, IP packet as a data pack is encapsulated in the Ethernet frame. In making of the Ethernet frame, destination address required in the frame must be the physical address of the destination node.

Physical address is the address provided by network interface card (NIC) of the node. This is the only address that the physical transport network can recognise for transporting the Ethernet frame to the desired destination. Hence, before making the Ethernet frame, physical destination address must be known.

Software in a node determines the next node or hop based on the IP addresses in the IP packet. To transfer the IP packet to next node or hop, the corresponding physical address should be known. The address resolution protocol does the function of obtaining the physical address for a given IP address. It is like a telephone number, say, 03324315145, to know about the physical locations, which in this case are Kolkata (033), Tollygaunge (243) and subscriber location (15145).

Primitive technique used by ARP is table look-up technique, in which a table known as ARP table or binding table is used for address resolution. The table contains an array of data, each array having a protocol address and corresponding physical (= hardware) address. It is exactly like a tele-